INCH-POUND
MIL-DTL-915/81A
22 August 2002
SUPERSEDING
MIL-C-915/81
18 June 1985

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, ONE TWISTED PAIR AWG-20 FOR SUBMARINE OUTBOARD USE ONLY (NOT FOR INBOARD USE), TYPE 1PR-A20E

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-915 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction

First	-	Conductor of high strength copper alloy (Phelps-Dodge #PD-135 or equal, silver coated, size AWG 20, class B stranding, (7 by 0.0126-inch wire).
Second	-	Insulation of ethylene tetrafluoroethylene (TEFZEL), with solid colors (see fourth). Nominal wall thickness 0.010 inch. Maximum overall diameter 0.060 inch.
Third	-	Insulation jacket of clear natural polyethylene in accordance with L-P-390, type II, class L, grade 3. Nominal wall thickness 0.010 inch. Maximum overall diameter 0.082 inch.
Fourth	-	Two insulated and jacketed conductors, one white, one red, twisted together with a lay of 1-1/4 ± 1/8 inch.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Fifth - Cable jacket of black polyethylene copolymer, applied so as to completely fill the pair valleys. Unless otherwise specified in the contract or order, the cable jacket material shall be Union Carbide Corp. #DFDA-0588, black #9865 or equal. Cable jacket shall be capable of being removed without incurring damage to the insulation jacket or the insulation. Nominal wall thickness 0.060 inch. Minimum wall thickness at any cross section 0.054 inch. Cable overall diameter shall be 0.285 ± .005 inch.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 20°C, maximum	11.1
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to water	2000
Insulation resistance - megohms/1000 feet, minimum	5000
Insulation dielectric. Test 100 percent of insulated conductors before application of insulation jacket. Test in accordance with MIL-W-81381, using a test voltage of 8.0 kV (peak), minimum.	
Group A:	
Visual and dimensional	No failure
Conductor (before application of insulation)	
Silver coating - thickness in micro inches, minimum. (Test in accordance with ASTM B 298)	40
Tensile strength - pounds, minimum	55.6
Elongation - percent, minimum	6.0
(Test tensile strength and elongation in accordance with FED-STD-228, method 3212, except as follows. Test not less than 3 specimens per each 1000 feet of conductor. Specimen length shall not be less than 20 inches. Tensile strength shall be in pounds. Specimen zero length shall be as near to 10 inches as practical.)	
Group B:	
Cold bending, cable - at minus 40°C, 3-inch diameter, mandrel.	No damage
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum	5000
Elongation - percent, minimum	150
Insulation jacket	
Tensile strength - lbf/in^2, minimum	1400
Elongation - percent, minimum	300

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Cable jacket
Tensile strength - lbf/in^2, minimum 1800
Elongation - percent, minimum 500

Group C:
Physicals (aged)
Insulation
Air oven (168 hours at 180°C)
Tensile strength - percent of unaged, minimum 90
Elongation - percent of unaged, minimum 90
Insulation jacket
Air oven (96 hours at 100 ± 2°C)
Elongation - percent of original 60
Cable jacket
Air oven (96 hours at 100 ± 20°C)
This test shall be in accordance with
MIL-DTL-915 ..... No failure
Environmental stress crack
This test shall be performed in accordance with
MIL-DTL-915 ..... No failure
Mutual capacitance - pF/ft at 2 kHz, maximum 25

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, plus all of groups A, B and C.

UNIT ORDERING LENGTH:
1000 feet (nominal).

PART OR IDENTIFYING NUMBER (PIN): The PIN consists of the letter M, the general specification number along with the specification sheet, and a dash number representing type and size. Refer to MIL-DTL-915 for an example.

M	**915**	**/81-**	**X-X**
Prefix for Military Specification	Specification Number	Specification Sheet Number	Type and Size

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Custodians:
Army – MI
Navy – SH

Review activities:
Army – AR, CR
Navy - CG, EC
DLA - CC

Preparing activity:
Navy - SH
(Project 6145-2317-013)

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